Vanilla Cultivation and Processing in Sri Lanka

A Comprehensive Technical Guide to Cultivation, Curing, and Processing

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1. Introduction

Vanilla (Vanilla fragrans, Vanilla planifolia) is considered one of the finest flavorings and aromatics used globally to sweeten, flavor, and enhance the appeal of foods consumed by humans, and is ranked among spices as second in price only to saffron. It is produced from pods that grow on vanilla vines. The word ‘vanilla’ originates from Spanish: ‘vaina’ means pod, and ‘illa’ means small, conveying the meaning of ‘small pod’.



Figure 1. Cured vanilla beans (Vanilla planifolia), a valuable natural source of flavoring and aroma

Although there is a huge demand for natural vanilla in the world market, countries cultivating the plant have failed to supply sufficient quantities. Furthermore, with the advancement of chemistry, artificially produced vanilla has also entered the world market.



Figure 2. Flower of the vanilla orchid

Since it was first introduced to the world in the 16th century, vanilla has maintained the coveted distinction of being the world's most popular spice. It is beautifully scented, filled with flavor, and adaptable enough to be used for just about every confection.



Figure 3. Cured vanilla beans and extracted vanilla seeds used for natural flavoring

2. Origin and Distribution

The vanilla plant is believed to have originated in northeastern Mexico, Guatemala, and other regions of Central America. Records of vanilla cultivation date back to before the 15th century; Mexicans cultivated vanilla for their own use for at least a thousand years before Europeans discovered America. In the 16th century a huge demand for vanilla arose, and records indicate that vanilla-flavored chocolate production also began during this period.

The vanilla crop, which naturally spread across forests, began as systematic cultivation in Java in 1846 and subsequently spread to Asian countries such as the Seychelles, Mauritius, Tahiti, Uganda, Comoros, and India.

There are no clear records of the exact date vanilla cultivation began in Sri Lanka. Currently, vanilla cultivation in Sri Lanka is distributed across the following districts:

- Kandy
- Matale
- Kegalle
- Nuwara Eliya
- Rathnapura
- Kurunegala

Denmark, the USA, France, and Canada are among the major consuming countries, while Indonesia, Madagascar, Mexico, Uganda, and India are the major suppliers. Indonesia is the major supplier by volume, while Madagascar supplies the highest quality vanilla. According to FAO estimates, vanilla is cultivated across approximately 45,000 hectares worldwide, with Madagascar and Indonesia recognized as the main vanilla-cultivating countries. In addition to Sri Lanka, countries such as the Comoros Islands, the Philippines, Java, Papua New Guinea, Fiji, Kenya, and Zimbabwe also cultivate vanilla. China has recently entered vanilla cultivation, growing it as a household crop within polytunnels. As global vanilla consumption shows an annual growth rate of 7%, the vanilla industry has a bright future. Although vanilla expanded as an economic crop during 1930-1940, this was subdued by the use of artificial vanilla, but it has recently been expanding again as a crop of high economic value.

2.1 History of Vanilla

Vanilla planifolia is native to Mexico and may have been utilized as a flavoring by the Totonac culture up to 1000 years ago. Because vines were considered a gift from the gods, the Totonacs cultivated them with almost religious zeal, often growing them around their homes

and watering them daily as a prized asset. Vanilla beans were also sent to the Aztec Emperor as a gift.



Figure 4. Historic Aztec-era vanilla illustration

While searching for treasures in the New World, the Spanish conquistador Hernan Cortes met Emperor Montezuma in 1518 and observed the Emperor drinking Chocolatl, a regal beverage made of vanilla-scented chocolate. Cortes brought sacks of cocoa and vanilla back to Spain along with treasure from Montezuma's empire, and within half a century chocolate flavored with vanilla was being produced in Spanish facilities. For a long time, vanilla was used by Europeans solely in conjunction with cocoa.

By 1602, vanilla was being used as a stand-alone flavor, according to Queen Elizabeth's apothecary, Hugh Morgan. Vanilla's popularity then increased beyond that of chocolate and any other flavor known before or since. For more than 300 years after Cortez discovered it, vanilla was produced exclusively in Mexico.

Vanilla plants were grown in various other countries, but the orchids never produced fruit until 1836, when the Belgian botanist Charles Morren discovered that ordinary insects could not pollinate the orchid. He found that the Melipone, a small bee found exclusively in Mexico's vanilla-growing areas, is uniquely able to pollinate the blossoms. Because the bee could not survive outside Mexico, Morren devised a method of hand-pollinating vanilla flowers. Later, the vanilla bean was sent to Reunion, a small island off Madagascar, by the

French, and Edmond Albius, a former slave, invented a rapid, easy method of hand-pollination that is still used today.

The French began cultivating vanilla on several of their islands in the Indian Ocean, the East and West Indies, and Oceania soon after this discovery; the Dutch planted it in Indonesia; and the British carried it to southern India. The push for extensive cultivation in the Indian Ocean region followed the invention of hand pollination. Madagascar, Comoros, and Reunion today account for 75% of current global output. Scientists continue to work on enhancing vanilla flavor and propagating plants through tissue culture.

Vanilla is also regarded in folklore as one of the oldest aphrodisiacs. According to legend, Xanat, the youthful daughter of a Mexican fertility goddess, fell in love with a Totonac youth; unable to marry him due to her divine nature, she transformed herself into the vanilla orchid to remain forever connected to him and his people. A Vanilla Festival marking the end of harvest is still celebrated by Totonac communities with dances and feasts.



Figure 5. Vanilla extract poured from a bundle of cured beans - the end product of the historic curing traditions described above

3. Botanical Characteristics of the Vanilla Plant

The vanilla plant is a monocotyledonous plant belonging to the orchid family (Orchidaceae), the second largest botanical family, comprising over 700 genera and more than 20,000-35,000 species. Around 110 species of vanilla have been identified (Purseglove et al., 1981), but only three are economically important: Vanilla fragrans (also known as Vanilla planifolia), Vanilla

pompona, and Vanilla tahitensis. Among these, the most important and widely cultivated commercial species is Vanilla fragrans, or Vanilla planifolia - the only edible species among all orchids, and the only orchid whose fruit yields commercially significant flavor and fragrance compounds.

3.1 Taxonomic Classification

Rank	**Classification**
Kingdom	Vegetal
Type	Fanerogamae
Class	Monocotyledonae
Order	Microspermae
Family	Orchidaceae
Tribe	Vanilleae Blume
Genus	Vanilla Swartz
Species	Vanilla planifolia Andrews; Vanilla pompona Schiede; Vanilla tahitensis J.W. Moore

3.2 The Vanilla Vine

The vanilla vine is a fleshy, green-leaved, herbaceous perennial that climbs upwards with the support of a host plant, possessing an axillary bud at each node. Composed of weak, succulent, flexuous, and brittle green stems that are cylindrical, monopodial, and simple or branching (1-2 cm in diameter, with internodes 5-15 cm long), the vine can climb about 30-35 feet along its support if allowed to grow freely, though under cultivation it is trained to a manageable height for hand pollination and harvesting.

Leaves alternate at every node. A leaf is generally 6-7 inches long and about 2-2½ inches wide, subsessile, flat, succulent, oblong-elliptic to lanceolate, with a rounded base and an acuminate tip, and numerous parallel veins; an axillary bud emerges from each leaf axil and develops into either a vegetative bud or a flower bud.



Figure 6. Vanilla foliage and climbing stems on a support tree, showing the alternating leaf arrangement and vine habit typical of the species.

Elongated whitish adventitious roots, about ¼ inch(6.35 mm) in circumference, emerge singly opposite the leaves at each node and firmly cling to the support; at the base they ramify within the humus or mulch layer, and the plant has endotrophic mycorrhizal association, with an underdeveloped outer parchment sheath (velamen). Additionally, hanging aerial roots absorb atmospheric water vapor, giving the vine good drought tolerance.

3.3 Flowers

Flowers bloom mainly from April to May, with a secondary flowering period in September-October. They open early in the morning and are greenish-yellow in colour, borne on robust axillary racemes of about 30-35 flowers each, blooming successively with usually only one to three flowers open at a time, and each flower lasting a single day (the raceme's full blooming

phase averages about 24 days). Flowers are bisexual; the perianth is formed by united sepals and petals, six in total. Five petals are uniformly arranged externally, while the middle petal is modified into a funnel- or trumpet-shaped structure called the labellum (lip). The fleshy column within this structure houses the gynostemium (fused pistil and stamen). The anther sits at the top of the column, divided into two pollen-bearing lobes, and hangs over the stigma but is separated from it by a flap-like structure called the rostellum. The stigma has three parts: one sterile (the rostellum/tip) and two fertile parts combined into the functional stigma. The ovary is inferior, with three carpels, forming a loculicidal capsule.



Figure 7. An opening vanilla flower with developing green pods beneath it on the same raceme.

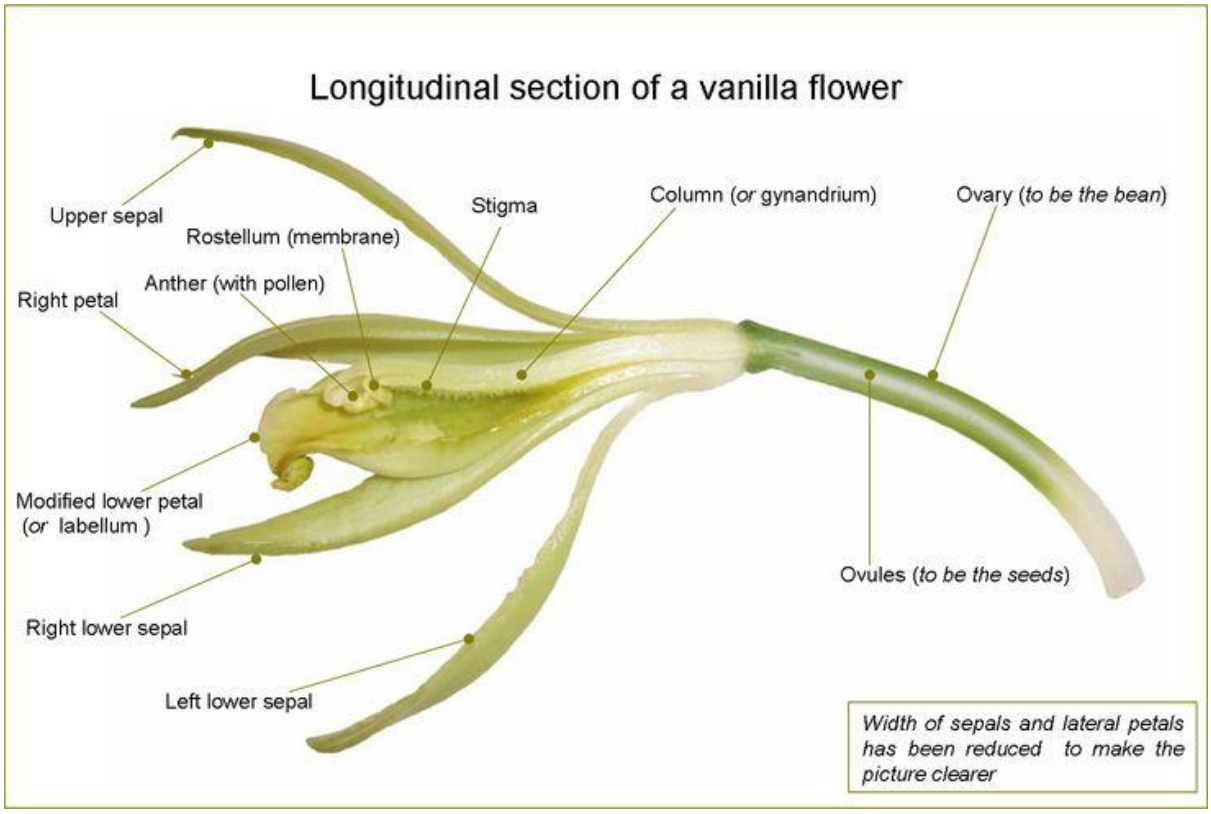


Figure 8. Labelled longitudinal section of a vanilla flower, showing the sepals, petals, labellum, column, rostellum, anther, stigma, and ovary

3.4 The Vanilla Pod

Successfully pollinated flowers develop into pods (capsules, commercially called beans). It takes about 7-9 months for a pod to mature, varying with prevailing weather conditions. A mature pod is pendulous, narrowly cylindrical and obscurely three-angled, generally about 8-9 inches long with a diameter of about ½-¾ inch. When fully mature it splits lengthwise and releases myriads of extremely tiny seeds, each about 0.003 inch in diameter. Under natural conditions seed germination is very rare, but artificial seedlings can be obtained using tissue culture media. In commercial cultivation, capsules are harvested before they are fully mature and before splitting occurs; in some vanilla-producing countries, owners prick distinctive marks on the green pods on the vine to deter theft.



Figure 9. Young, unripe green vanilla pods hanging in a cluster on the vine, prior to maturation.

3.5 Vanilla Cultivars

Bourbon vanilla, also called Bourbon-Madagascar vanilla, is grown across the Indian Ocean islands (Madagascar, Comoros, Reunion - historically the Île Bourbon). The term also describes the characteristic flavor produced by V. planifolia, the tropical species cultivated effectively in Sri Lanka and India. Its flavor is typical, with smooth, creamy finish notes and a full aftertaste of dried-fruit, cinnamon-like essence, and a pleasant, well-balanced aroma considered the most complete profile of any vanilla extract.

Tahitian vanilla (Vanilla tahitensis), genetically believed to be a hybrid of V. planifolia and V. odorata, is grown only on Tahiti and Moorea. Its flavor releases quickly, is rather sweet, with mild, somewhat oily undertones.

Pink pompona (Vanilla pompona) is grown in the Caribbean, Central America, and South America; it is used mainly in pharmaceuticals and perfumery rather than as a food flavoring.

4. Climate Required for Vanilla Cultivation

In the wild, Vanilla planifolia grows by climbing trees in damp tropical lowland forest, from sea level up to about 600 meters. It grows best in hot, humid, insular climates with frequent but not excessive rain, and areas with a prolonged dry season should be avoided for cultivation.

Key climatic and site requirements for successful vanilla cultivation are summarized below:

- Temperature: 21-32°C, with 27°C as the optimum. Dry weather and sea breezes are unsuitable.
- Rainfall: well-distributed rainfall of 2,000-2,500 mm over about 10 months. Wet and intermediate zones are suitable. A dry period of about 2 months is required to induce flowering.
- Elevation: sea level up to about 3,000 feet.
- Soil: very well-drained, light, friable, loamy soil rich in organic matter, on gently sloping terrain with a thick surface layer of humus or leaf mulch; pH 6 to 7 is preferred. Waterlogged land and paddy land are unsuitable.
- Shade: 70%-75% shade, kept consistent through the year, reducible to 30% when inducing flowering.

4.1 Temperature

This crop can be successfully cultivated in areas with temperatures between 21 and 32°C, with the most suitable (optimum) temperature being 27°C. Dry weather and sea breezes are not suitable.

4.2 Rainfall

Vanilla requires well-distributed rainfall over 10 months, ranging between 2,000 mm and 2,500 mm. Wet zones and intermediate zones are suitable. To induce flowering, subjecting the vines to stress is necessary, requiring a dry period of about 2 months.

4.3 Elevation

The crop can be successfully cultivated from sea level up to about 3,000 feet.

4.4 Soil Requirements

Very well-drained soil is suitable for vanilla cultivation. Sloping land is ideal, as water drains well on such terrain, and vanilla can be grown successfully on gently sloping terrain with light, friable, loamy soil rich in organic matter, with enough but not excessive drainage, and a thick surface layer of humus or leaf mulch to let the roots spread. Waterlogged land, where excessive water retention is dangerous to the crop, should not be chosen - vanilla cuttings planted in waterlogged soil rot, and paddy lands are particularly unsuitable. A soil pH range of 6 to 7 is preferred.

Before planting, soil testing can be used to assess pH, nutrient status, and organic matter content. Where drainage is poor, raised beds or drainage channels can be considered to reduce waterlogging.

4.5 Shade Requirements

Successful cultivation requires 70%-75% shade, which should be maintained consistently and balanced throughout the year, except when shade reduction is necessary. Adequate shade reduces moisture loss from the surrounding environment, protects organic matter during the dry months, and protects the beans from low temperatures during cold months. Too much shade causes stem weakening, while too little shade causes sunburn during the dry season; in either case, the plant is weakened and becomes vulnerable to disease. It should be possible to prune shade trees to reduce shade down to 30% when necessary (see section 11, Stressing the Vines).

Suitable shade species:

- Tall shade: Mara, Casuarina (Sauuru), and Erythrina (Erabudu).
- Lower-level shade: coffee, banana, Gliricidia, and Erythrina.

In Kerala state, India, Gliricidia as well as Erythrina plants are used as supports. Vanilla cultivation cannot be successfully undertaken under rubber, because rubber roots invade the vanilla mulch, absorb nutrients, and act detrimentally on vanilla vine growth.

4.6 Land Orientation and Site Selection

Where slopes prevail, east-facing sites may be preferred because they receive morning sunlight and are generally less exposed to intense afternoon sunlight, helping reduce soil moisture loss and the risk of sunburn to vanilla plants and pods.

- Consider wind exposure, accessibility for planting and harvesting, and any previous occurrence of serious pests or diseases at the site.
- Existing shade-loving vegetation or established trees can indicate a favorable local microclimate.

4.7 Irrigation

During dry periods, supplementary irrigation may be needed. Watering should maintain a moist root zone without waterlogging, with excess water able to drain freely. Watering frequency should be adjusted according to rainfall, soil moisture, and local microclimatic conditions.

5. Propagation

Vegetative propagation methods are used for propagating vanilla; there is no notable difference among currently grown vines. Stem cuttings are the standard commercial planting material, with a suitable cutting length of 3 to 9 feet - the amount of planting material available typically determines the length actually used.

Cuttings should be selected from vigorous, high-yielding mother vines with the following characteristics:

- Stems with large and fleshy leaves
- Free from pests and diseases
- Rapid growth habit
- High yield

For planting, a cutting with seven nodes and the apical shoot, or a stem matured to that size, is suitable. Cuttings with over-mature, very dark green stems take a long time to shoot and should not be preferred over vigorous planting material. Similarly, vine cuttings that have already flowered or from which pods have already been harvested are not suitable for planting.

Vanilla can also be multiplied through tissue culture; such plants have a longer lifespan than regular plants, and tissue for culture can be obtained from vines selected as mother plants.



Figure 10. Young vanilla plants raised in nursery bags, ready for field planting.

Disinfection of cuttings:

Young vanilla cuttings should be inspected for signs of disease before planting to reduce the risk of infections such as rot and anthracnose. The three basal leaves may be carefully removed using a clean knife or pruning shears to prepare the lower portion of the cutting for planting.



Figure 11. Newly planted vanilla cuttings established at the base of support stakes in the field.

6. Intercropping and Companion Planting with Vanilla

Vanilla can be cultivated with other perennial crops, provided these do not provide excessive shade. Intercropping vanilla with annual crops is not recommended, since the frequent soil loosening involved can damage the vanilla vine's roots. Likewise, using commercial crops as living supports should be avoided, as maintenance activities for those crops may harm the vanilla vines and shade cannot be properly controlled when stressing the vanilla vine. Vanilla cultivation can be done with coffee and banana, which increase shade and atmospheric moisture while also acting as a windbreak.

Suitable and unsuitable companions:

The suitability of companion plants depends on their growth characteristics and their compatibility with the cultivation requirements of vanilla.

- Companion plants should allow filtered light.
- They should not compete heavily with vanilla for water and nutrients.
- They should not introduce additional pest or disease pressure.
- Avoid sun-loving crops requiring full sun exposure.
- Avoid plants with aggressive root systems that compete heavily for water and nutrients.

- Avoid disease-prone crops that could increase disease pressure around vanilla.

For unfamiliar crop combinations, a small-scale trial is recommended first, to observe effects on shade, soil moisture, resource competition, and vanilla growth before wider planting.

7. Land Preparation for Vanilla Cultivation

Unwanted bush growth in the field should be cleared, and underbrush removed using a billhook (visikaththa). Subsequently, determine the slope and establish contour drains at appropriate distances to ensure proper drainage and minimize water stagnation in the field. Contour lines can be marked using an 'A-frame', and planting spots (peg marking) can be marked at the same time. Support stakes (shade/tutor trees, see section 4.5) must be planted at this stage, at least 6 months prior to planting the vines. Weeds should be cleared with a hoe in a 3-foot diameter around each support established in the field.



Figure 12. Vanilla vines established and growing along support trees (tutors) in a shaded plantation system.

7.1 Planting Spacing for Vanilla

Vanilla vines are planted at a spacing of 8 × 5 feet. This spacing is recommended for cultivations aimed at pod production, giving a plant requirement of about 1,089 vines per acre.

7.2 Land Preparation for Planting Vines

Around the support planted at 8 × 5 feet spacing, loosen the soil in a 3-foot diameter and ½-foot depth, then add leaf manure to this and mix well.

7.3 Pre-treatment of Vines

Take 3-foot long vine cuttings and remove about 3 leaves at their basal end. Cut and remove the stem very close to the lowest node, since the tissue is firmest at that spot, minimizing fungal damage. Afterward, hang the vine in a shaded place for about 5 to 6 days with its upper end pointing downward; the number of days vines are hung varies with climate and region, but care must be taken to plant the vine before it shrivels. Hanging causes water in the vines to evaporate, stimulates callus formation at the cut ends, and stimulates bud and root growth; this callus formation reduces the chance of damage by soil fungi.

7.4 Planting Vanilla Vines

Make a groove horizontally across from the base of the support to a depth of about 2½ inches. Plant a length of about 1½ feet (about 3 or 4 nodes) of the vanilla vine in the prepared groove. At this stage, the first ½ inch of the basal end of the vanilla vine should be left exposed and uncovered by soil to reduce the risk of infection by soil-borne pathogens. Carefully bend the vine towards the previously established support and tie it securely to keep it stable during establishment. The apical tip of the vine should then be attached to the support to guide its upward growth and encourage climbing. After positioning and securing the cutting, arrange dried banana leaves around the leaves and stem to provide protection from direct sunlight. This protective covering helps reduce excessive exposure to sunlight while the newly planted vine becomes established.



Figure 13. Planting a vanilla cutting at the base of a support tree, with a mulch ring of coconut husks around the stake

Planting time:

Under Kandy conditions, planting is generally favored around the beginning of the rainy season, commonly May to June, so that adequate moisture is available during establishment; planting during excessively hot or dry periods should be avoided.

- Newly planted vines should be monitored closely during establishment and given supplementary water when rainfall is insufficient.
- Organic mulch around the base helps conserve moisture.

- Temporary shade may be useful for young vines until sufficient foliage develops.



Figure 14. Recently established young vanilla plants with coconut-husk mulch rings around each support stake, shortly after planting.

7.5 Mulching and Nutrient Supply

A mulch of decomposed leaf litter should be applied to cover the base of the vine up to a height of 1½ feet, and renewed as it decomposes. Coconut husk mulch is rapidly attacked by termites, so constant vigilance is important.



Figure 15. Worker tending vanilla vines

8. Training the Vines

Remove lateral branches on the support plant from ground level up to 5 feet, and allow 3-4 lateral branches to develop from 5 feet upward. Vines are trained in two main ways: allowing them to grow upwards along the support and looping, or horizontal training.



Figure 16. Farmer inspecting a vanilla vine

8.1 Looping

After the vanilla vine climbing along the support reaches a height of about 5½ feet and has grown 2-3 feet along the upper branches of the support, carefully detach it, wind it around a branch of the support, and guide it downward. When the trailing vine reaches ground level, remove the leaves on the part touching the soil, bury that part in the mulch, and tie its base to the support - this forms a loop.

- It is advisable to make 4 or 5 such loops per vine.
- Making loops creates several new root systems from the nodes.
- Looping turns the vine downward and keeps it at an accessible height for management.
- Looping also stimulates flowering.

Looping (provignage) as a vine-renewal practice:

Looping also serves to renew the growth capacity of older vines. A mature vine segment is brought back toward the growing substrate to improve the plant's nutritional capacity through additional root development. The upper part of the loop can produce a new growing head from the axillary bud below it. This new vine may later be layered again to increase plant mass or managed as a downward-growing productive vine (bout-pendant).

Important precautions before looping:

- Use a vine segment that is sufficiently mature (at least 3-4 weeks old) for the portion returned to the substrate, since immature tissue is more susceptible to rot and disease.

- Remove the leaves and exhausted roots from the portion that will be placed in the substrate.
- Allow the cut surface to heal before returning it to the substrate.

Suggested vine lengths for looping:

- First looping of a small cutting: approximately 1.2 m.
- Second looping of a small cutting: approximately 2.4 m.
- Looping of an adult vine already in production: approximately 4.5-5 m.

Looping must be balanced, as insufficient looping can weaken the vine, while excessive looping can reduce flowering and productivity. The sustainable vanilla cultivation guide for Madagascar recommends performing looping approximately 3-4 times per year, depending on vine growth and management requirements. Old or senescent vine portions should also be removed so that younger tissue can continue developing. A vine segment may become senescent after about three years, while a downward-growing productive vine may lose productivity after two successive flowerings.

Looping is not a natural phenomenon of vanilla plants. Rather, it is an agricultural activity to encourage them to live longer and be more productive. Mastering this operation is therefore one of the keys to the success of vanilla cultivation.

Vanilla vine configuration can be used to extend the planted area:

▪ In the narrowest definition, looping makes it possible to maintain the plant mass at the height of a man by lowering and bending up the high vines on the same stake or on an overhead support between stakes without necessarily putting the apical end back into the substrate.



Figure 17. Developing pods on a looped vine

8.2 Horizontal Training of Vines

Vines planted at 8 × 5 feet spacing can also be trained horizontally by running them spirally on supports; a thick nylon cord, or a branch of the support carefully bent downward, can be used for this. Four or five loops can be made between two supports on this bent branch; as before, leaves are removed where the loop touches the soil, and that part is buried in the mulch. Damage to vines by wind is higher with this method, and disease also spreads more rapidly.

9. Fertilizer Application

Well-decomposed compost fertilizer is generally applied, with the leaf mulch continuously renewed. Gliricidia leaves and wild sunflower leaves, which are rich in nitrogen, can be used for this purpose. Well-decomposed farmyard manure can also be used to increase soil organic matter, while fresh manure should be avoided around young vines. A surface layer of organic mulch helps conserve soil moisture, suppress weed growth, and gradually improve soil organic matter while supplying nutrients as it decomposes. Therefore, the mulch should be regularly renewed to maintain a steady nutrient supply to the vanilla vine.

10. Pests and Diseases

Although insect pests causing economic damage to vanilla cultivation are rarely observed, several diseases have been reported, among which Phytophthora rot occupies a primary place. Diseases can be prevented by maintaining cleanliness in the cultivation field under proper management and by following cultivation instructions. Vanilla is commonly grown in humid environments under rain-fed conditions, making it susceptible to fungal and other diseases from the surrounding tree canopy and biomass.

Unsuitable growing conditions that promote the development of most diseases include:

- Overwatering and improper drainage
- Excessively thick mulch
- Over-pollination
- Too much shade

General best practices to control diseases:

- Keep plant nurseries with a high population of plants clean.
- Use suitable mulch, and check compost or leaf mold thoroughly before use.
- Check for excess water and waterlogging in irrigation systems, since these can cause fungal diseases.
- Ensure proper ventilation and air circulation in dense cultivation systems.
- Avoid excessive organic manure, excessive mulch, and excessive watering.
- Cut and burn off disease-infected plant portions.
- Do not transport branch cuttings or planting material from affected regions.
- Avoid planting too close together or too densely, and avoid excessive shade.
- Visit vanilla gardens regularly to detect disease outbreaks early and act promptly.



Figure 18. A vanilla vine showing disease symptoms

10.1 Phytophthora Rot (Bean Rot / Stem Blight)

Caused by Phytophthora medii (Phytophthora meadii). Spread is favored by excessive shade, vine overcrowding, waterlogged conditions, constant rain, and presence of Phytophthora inoculum.

Symptoms:

- First starts at the base of the pod, where the tip turns black.
- Spreads upward on the pod as a black, wet rot.
- Eventually spreads to stem and leaves, appearing as brown patches.
- Can progress to blighted areas several centimeters down the stem.
- Eventually the entire vine rots and dies, including rotting of the apical shoot.



Figure 19. Pods affected by bean rot



Figure 20. A vanilla pod with a dark, necrotic patch near the tip, characteristic of Phytophthora bean rot.

Disease control:

Cut off and remove infected parts, and apply a 1% Bordeaux mixture (or 0.2% copper oxychloride) to the remaining crop. Reduce excessive shade and overcrowding to improve air circulation.

10.2 Sooty Mold Fungus Damage

The tip of the vanilla leaf appears withered; this gradually develops and kills the leaves. Promptly prune and remove heavily affected or dead leaves to prevent the spread of the issue.



Figure 21. Foliage with fungal leaf damage

10.3 Fusarium Fungus Damage (Root Rot / Leaf Rot)

Caused by Fusarium oxysporum / Fusarium batatis var. vanillae, this damage is typically found at the point where the stem joins the soil, and is most severe at environmental temperatures between 14 and 32°C; high atmospheric humidity increases the damage.

Symptoms:

- Leaf tip turns yellowish to brown, and rotting progresses toward the leaf center.
- In the root system, browning begins at the root tips, which degrade and eventually die.

- Aerial roots are also affected.
- Infected plants become flaccid, with shrinking stems and drooping vines, especially during dry, sunny periods when water and mineral uptake is reduced.

Vanilla planifolia is the most vulnerable of the vanilla species to this disease.



Figure 22. Exposed vanilla root system at the base of a support tree, showing root condition affected by soil-borne fungal pathogens.

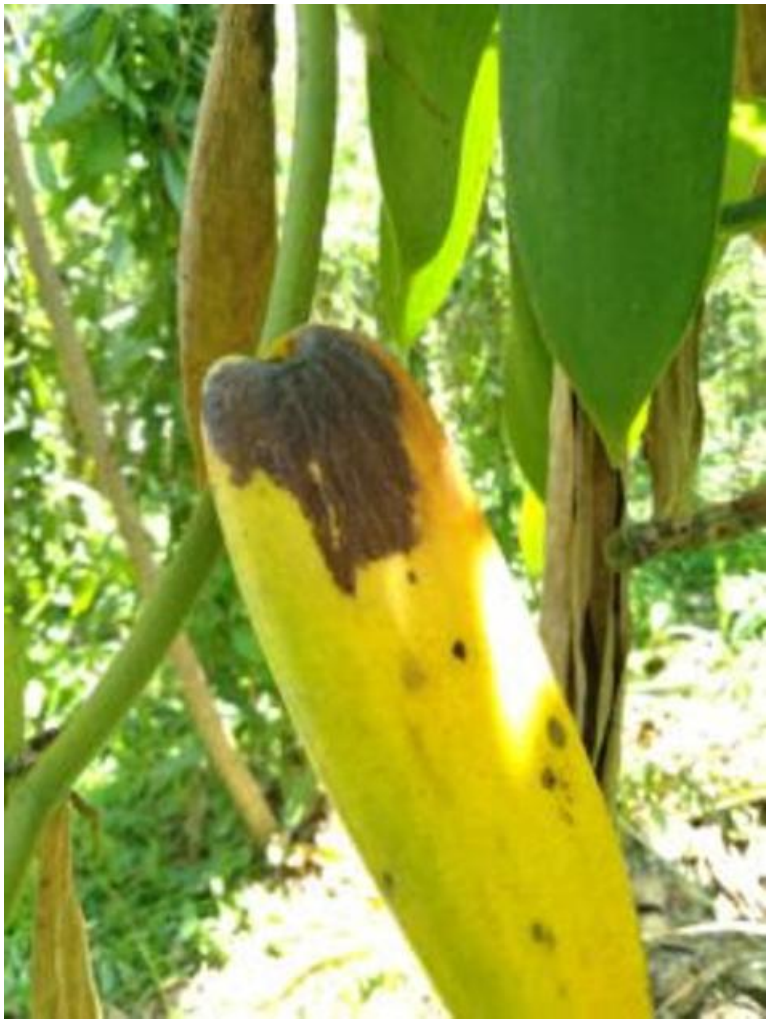


Figure 23. A leaf with fungal rot symptoms

Disease control:

Applying Bordeaux mixture (1%) has given good results; alternatively, 0.2% carbendazim, 0.2% Mancozeb, or soil-soaking with 0.2% carbendazim or 0.2% copper oxychloride can be used. Biocontrol agents such as Trichoderma viride, T. harzianum, and Pseudomonas fluorescens have been found to inhibit Fusarium oxysporum and Sclerotium rolfsii under laboratory conditions; applying Trichoderma at the base of vines can reduce inoculum build-up and root rot incidence.

10.4 Other Fungal Diseases

Anthracnose (Massae Calospora vanillae):

Reasons for spreading: poor drainage, excessive soil moisture, extended wet weather, vine overcrowding, sick and dead leaves, beans, and aerial roots that fall to the ground.

Symptoms: The first signs of infection appear on fragile portions such as immature beans and the tips of aerial roots. The symptoms emerge on the beans as exceedingly minute pale red or amber-cultured pustules that grow up in tiny groups to produce discolored areas. They become black at the tip and in the center. Then they wilt and fall off after 2-3 days. Although the pustules occur on both sides of the leaves, the severity is greater on the top side. As the illness advances, the leaves begin to yellow and dry, eventually dying.

To prevent the illness from spreading further, proper phytosanitation (collection and burning of infected areas) is required.



Figure 24. A leaf showing Anthracnose disease symptoms

Inflorescence rot (Colletotrichum sp.):

Causes flower buds to wither, beginning as brown spots on the flower stalk during elongation of the inflorescence, which progressively spreads to the buds and causes them to shed; spraying flower buds with 0.2% carbendazim effectively limits spread.



Figure 25. A vanilla flower bud and developing inflorescence, the tissue most vulnerable to inflorescence rot and related fungal infections.

Shoot-tip rot (Colletotrichum gloeosporioides, Fusarium oxysporum):

Begins as brown spots on the petiole and lower regions of the youngest, barely-unfolded leaf, whose funnel shape traps rainwater or dew, creating ideal rotting conditions; within a few days the rot spreads to affected leaves and the shoot tip softens, browns, and falls off, with the infection generally confined to the shoot tips. A 0.2% carbendazim spray is used as control.

Stem rot:

Presents as water-soaked, brown sores spreading to both sides of the stem, later forming elongated areas of tissue degradation; leaves yellow and dry out in advanced stages. If the stem becomes diseased and shriveled in the basal or central regions, the distal portions wilt unless sufficient aerial roots have established in the soil; stem infection is sometimes followed by bean rotting.

Viral diseases:

Commercially grown vanilla is also affected by Cymbidium Mosaic Virus (CyMV, usually causing only mild mottling or chlorotic streaks, often asymptomatic), Odontoglossum Ringspot Virus (ORSV, typically asymptomatic or causing mild mottling, spread by sap and planting-material cuttings), and Vanilla Necrosis Potyvirus, the most severe viral disease affecting the crop, which causes widespread chlorotic spotting and leaf deformation in young leaves, necrotic lesions on older leaves and stems, and, in advanced stages, death of affected vines.

Viral disease management:

- Viral-infected vines should be removed and burned or deeply buried.
- Planting material should never be taken from mosaic-affected vines.
- Hands or tools that have contacted a diseased plant should be thoroughly cleaned before touching a healthy vine.

11. Stressing the Vines

Stressing the vanilla vine is essential to stimulate flowering. About 4 months before flowering, in December, shade should be pruned so that shade is reduced to 30% and light reception increased to 70%. This is done by cutting and removing branches of the support plant.

- 7 nodes are cut and removed from the apical portion of hanging vines, destroying apical dominance and stimulating lateral buds.
- Looped vines are not pruned at this stage.
- Subjecting the vine to stress in this manner is mandatory for flowering to occur.

12. Flowering

Flowers begin to emerge about 4 months after the vines are stressed. Flowering starts in April and continues daily for about a month, with a secondary period possible in September-October depending on regional weather conditions. Vanilla typically begins flowering in its second or third year after planting, depending on the size of the initial cuttings, reaching peak flowering productivity at 7 to 8 years. The yellowish-green flowers are borne on racemes containing about 30 flowers each, with 2-3 flowers blooming per day. Each flower opens early in the morning, remains receptive for approximately eight hours, and lasts only one day before closing permanently; if not pollinated, it generally falls the following day. Overall, approximately 200-300 flowers may bloom on a single vine annually.

For commercial cultivation, artificial hand pollination is necessary because natural insect pollination is very rare. Although vanilla flowers are self-fertile, the rostellum separates the anther from the stigma, preventing spontaneous self-pollination. Pollination therefore requires transferring pollen to the stigma or lifting the rostellum and pressing the anther against the stigma. The most suitable time for hand pollination is between 7:00 AM and 11:00 AM, as flower receptivity decreases towards the afternoon. Pollination success is also reduced during periods of low humidity and intense dryness, whereas conditions following rainfall generally favor successful pollination.

To obtain straight, high-quality beans, flowers on the lower side of the raceme are preferably selected for pollination, as those on the upper side may produce crooked beans. Approximately 10-15 pods per cluster may be maintained, with about 100-150 pods per vine considered sufficient for productive growth. Hand pollination should be carried out on newly opened flowers throughout the flowering period until the desired number of pods has been established. Once the required number has formed, remaining buds can be removed by trimming the tip of the inflorescence. Two months after pollination, developing pods measuring less than 4 inches should be removed, along with damaged or deformed capsules, to encourage the remaining pods to develop properly and produce first-grade vanilla beans. Successfully developed pods generally take about 8-9 months after pollination to mature. Hand pollination is a labor-intensive activity and is estimated to account for approximately half of the total labor cost of commercial vanilla production.

12.1 Flower Pollination Process and Its Stages

The pistil and stamen are located within the column at the center of the flower, differentiated from the other petals. Hand pollination is carried out as follows in 3 steps:

1. Expose the column: Take a coconut leaf midrib (or any other fine stick, or a toothpick) with a sharp point, about 4 inches long. Holding the opened vanilla flower with the left hand, push the labellum (lip) downward using the thumb of the right hand. This exposes the column.

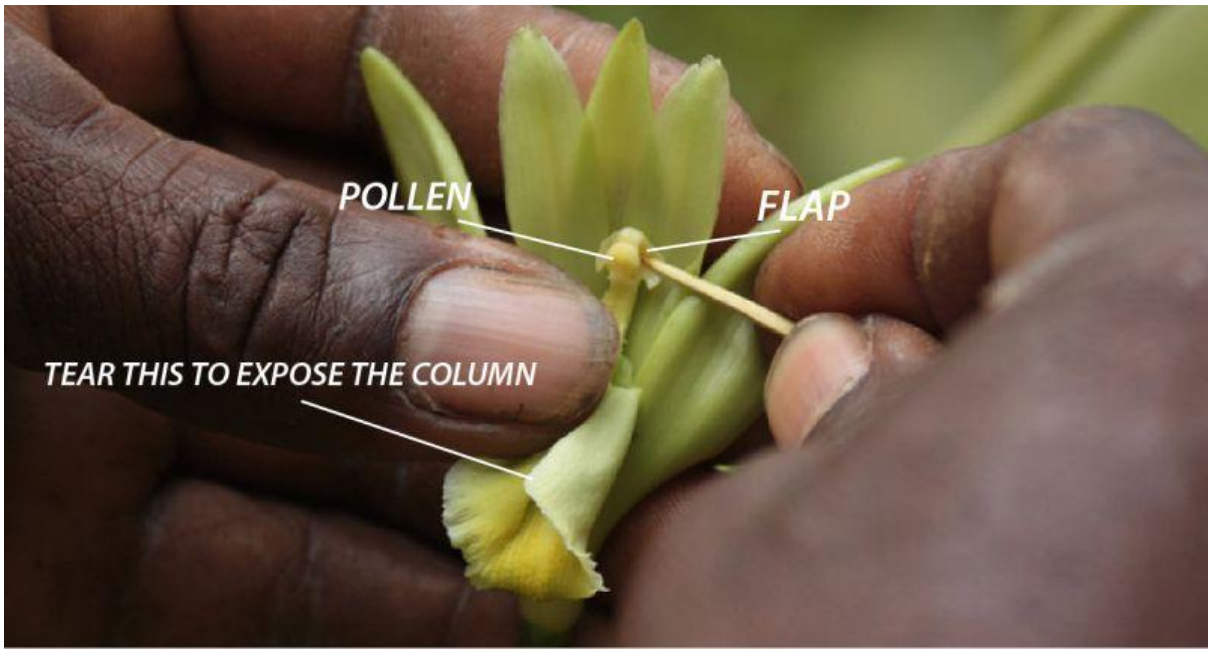


Figure 26. Exposing the flower column

2. Lift the rostellum: Using the coconut midrib (or the free end of the toothpick) in the right hand, gently lift the rostellum upward - the flap that covers the stamens and separates them from the stigma. This uncovers the stigma so it is clearly visible, while the rostellum is held back with the fingertips.
3. Transfer the pollen: Using the thumb of the left hand, lower the anther onto the stigma and press it gently, or carefully twist the toothpick to apply the pollen to the spherical stigma hole. Afterward, carefully remove the stick - the pollen grains are deposited on the stigma, completing pollination.



Figure 27. Hand-pollinating a vanilla flower - the labellum is held back to expose the column before pollen transfer.

If pollination was successful, the drooping flower remains attached to the plant and shows signs of pod formation after a week or two: the fertilized ovary enlarges and transforms into a fruit, and the ovules inside harden to form the seeds within it.

13. Harvesting

A pod resulting from a flower pollinated in April reaches harvesting stage around October to December. The seed pod develops over a period of about 7-9 months, and a healthy vine may yield up to 100 pods a year. Harvesting should be done when about 0.1 cm of the tip of the pod turns light yellow, before the pod becomes over-mature and splits. Harvesting is performed manually, with mature pods carefully detached from the cluster and collected in bags to minimize physical damage.



Figure 28. A cluster of mature green vanilla pods, developed and ready to approach harvest maturity.

Indicators that a pod is ready to harvest:

- About 0.1 cm of the tip of the pod turns light yellow.
- The thickest part of the fruit (the ‘flower end’) turns light yellow.
- The overall colour of the pod shifts from dark green to light green.
- Two distinct lines become visible along the length of the pod.
- Well-matured pods separate easily from the cluster when gently pulled or pushed sideways.

Harvesting at the correct maturity stage is essential for producing high-quality vanilla beans. Immature pods fail to develop the desired aroma and colour during processing, resulting in lower-quality vanilla products, while over-mature pods may split naturally or during processing, resulting in aroma loss, increased susceptibility to fungal infection, and reduced commercial quality. Approximately 60-70 well-matured green vanilla pods are required to produce 1 kg of dried vanilla beans. To reduce harvesting labor, a pod situated immediately next to a fully mature pod may also be harvested at the same time, provided it has reached an appropriate stage of maturity. Care should be taken to avoid damaging the remaining pods and clusters.



Figure 29. A bulk collection of harvested green vanilla pods, gathered immediately after harvesting prior to sorting.

13.1 Pruning Vines After Harvest

After harvesting is completed, branches from which pods were harvested will not flower again, so cutting and removing old, spent vines is a necessary task; otherwise these branches generate excessive vegetative growth, which reduces sunlight penetration and decreases pod setting.

With 1,000 vines per acre, approximate dry-pod yields are about 130 lbs. in the third year, 300 lbs. in the fourth year, and about 600 lbs. in the fifth year. These yields can be sustained for up to 15 years, after which yields gradually decrease and replanting should be undertaken.

14. Factors to Consider for High-Quality Final Production

Pods formed after flowering should be harvested within 8-9 months. High-quality vanilla can only be achieved from well-matured pods, so it is advisable to select only well-matured pods for processing. Since vanilla pods in a cluster mature individually over 15-25 days, harvesting must be done individually and should not be delayed at any cost; the harvesting season should end within 8-10 weeks.

- During the main harvesting period, harvest twice a week.
- During the remainder of the season, harvest once a week.
- As pods mature, approximately 1 cm of the lower end turns yellow, indicating that they are ready for harvesting. Well-matured pods also separate easily from the cluster when gently pushed sideways.
- Harvesting a pod immediately adjacent to a well-matured pod at the same time can reduce labor costs, provided the adjacent pod has reached a suitable stage of maturity
- Neither immature pods nor pods split from over-maturity can yield a high-quality product.
- Care should be taken when handling large quantities of harvested pods, since overloading leads to bruising.

14.1 Sorting Raw Pods

Raw pods are sorted according to their size, quality, and degree of maturity. Grading before processing is important, as it provides uniform conditions that make it easier to produce a high-quality product.

Immature Pods:

Dark green and firm; these should not be harvested. Also, the tip of the pod has not changed color. Sometimes, depending on the amount of sunlight received, these may also be light green-yellow, but if the lower end of the pod is not yellow or brownish, it is not sufficiently mature. These pods have high moisture content. About 6 kg of these pods are required to make a kilogram of dry weight. If harvested under proper management, the proportion of immature pods broken can be minimized to 5%. These yield a low-quality product with low vanillin content

Over-Mature Pods:

These are yellowish pods split from the lower end upward. The tip takes a brown or black color and is not very firm. Less than 4.5-5 kg of raw pods are required to make a kilogram of dry pods. These also have a mild aroma. After drying, although they have the exterior appearance of good dry pods, they are lower in quality because they are split, but they have a higher vanillin content.

Mature vanilla pods ready for harvesting:

This pod should be taken. Yellowish-green color. Pods are firm. The tip of the pod being yellow is a sign of good maturity. When processed, a very high-quality product can be obtained, and the vanillin content is also high. 5 kg of raw pods are required for a dry kilogram. If vanilla yield is large, sorting and processing further is most appropriate.

Well-Matured Pods:

Plump, stem and dark color starting to turn yellow, non-splitting, over 12 cm long pods.

Over-Matured Pods:

Dark yellow, black, plump dark color with a slightly thin pod.

Short Pods:

Well-matured pods between 10-12 cm in length.

Immature Pods:

Plump, but no color change and age less than 8-9 months.

Weak Pods:

Light green, weak, flexible pods.

Very Short Pods:

Very small pods less than 10 cm long.

Rotted Pods:

Dark gray patches, unpleasant odor, mostly shriveled and light green. May leak juice. Since these are unsuitable for processing, discarding them is recommended. If a small part is rotted, that part can be cut off and the remaining part taken for separate processing.

15. Preparing Vanilla for the Market

To produce high-grade vanilla, well-matured pods must be harvested and processed as quickly as possible, ideally beginning within a week of harvest. The objectives of vanilla processing (curing) are:

- To develop the attractive, pleasant aroma of vanilla, since a raw pod has no pleasant aroma whatsoever.
- To achieve an appealing dark appearance in place of the green colour of raw pods.
- To process the pods into a stable, storable form.

During the last few days of ripening, the pod accumulates a glucoside, glucovanillin, which is broken down into glucose and free vanillin - the chemical primarily responsible for aroma - during the curing (fermentation) process. Curing a green vanilla pod has no market value unless the full curing procedure below is followed.



Figure 30. Sorting freshly harvested pods

Grading Green Vanilla Pods Before Curing:

Before curing, harvested green vanilla pods may be classified according to their length and physical condition. This classification helps separate pods of similar size for processing and is different from the final quality grading of cured vanilla pods.

- Grade A: length more than 15 cm.
- Grade B: 10-15 cm.
- Grade C: less than 10 cm.
- A separate category for cut, split, bruised, or otherwise damaged green beans.



Figure 31. Measuring pod length with a ruler

Graded beans should be washed and cleaned well before curing, to remove any residual chemical or Bordeaux-mixture stains - concentrated tamarind water or sour milk can also be used for this cleaning step.



Figure 32. Washing green vanilla pods

15.1 Primary Drying

This is a very important stage in vanilla processing, consisting of three sub-stages:

1. Killing the pods
2. Sweating (rapid drying)
3. Sun drying

If not done very carefully, severe damage can be caused to the pods during this process; prolonged rain is also unfavorable.

15.2 Secondary Drying

This stage comprises the following sub-steps:

1. Shade drying
2. Weighing and grading
3. Conditioning in boxes

15.3 Packing

This refers to packing the conditioned pods in a suitable manner for presentation to the market.

16. Main Processing Methods for Vanilla

Vanilla pod processing is carried out in several ways across the world, with two main methods generally followed: the Bourbon Method and the Mexican Method. Taken as a whole, the process of bringing vanilla pods to market can be described in four basic stages, which all methods share:

1. Killing or wilting stage
2. Rapid drying stage (sweating)
3. Slow drying
4. Conditioning (sweating-in-storage)

16.1 Killing or Wilting Stage

A step that must be very carefully and cautiously controlled during processing: it stops further growth of the pods, prevents splitting, and starts the process of creating pleasant aromatics inside the pod. A brown colour developing in the bean indicates it has been successfully killed.

16.2 Rapid Drying Stage (Sweating)

Killed pods are kept at a raised temperature in this step, stopping unwanted fermentation and promoting the necessary chemical changes. The removal of water from the pods here is similar to perspiration and is therefore also called sweating.

16.3 Slow Drying

Reduces the pods' weight to about one-third of their original weight while developing a soft texture, during which vanillin content continues to increase.

16.4 Conditioning

Pods are kept in airtight, closed boxes for about three months to bring them to the condition where the desired fragrance and characteristic flavor are permanently developed.

17. The Bourbon Method

In Sri Lanka, vanilla cultivation and processing take place mainly at higher elevations, where the curing process followed is similar to the Bourbon method used in Madagascar. After sorting raw pods by maturity and length, they must be immersed in hot water for killing. A large barrel cut down to approximately three-quarters of its original height, or another suitable container, can be used for the hot water treatment. A wire-mesh basket that fits inside the container is recommended for immersing the vanilla pods in hot water. A thermometer is also necessary to monitor the water temperature throughout the treatment.

17.1 Killing or Wilting Stage - Hot Water Treatment

After harvesting, pods are sorted and subjected to hot water treatment. Fill a barrel with water and heat it; set up the hearth and surroundings to minimize smoke. A thermometer is essential, since measuring water temperature accurately is important.

- Maintain the hot water at 60-65°C (heat it to slightly higher, about 70°C, to allow for cooling when pods are added).
- Very good pods: about 3 minutes immersion.
- Immature/long pods: 3-3½ minutes immersion.
- Short/defective/weak pods: 2½ minutes immersion.
- Water temperature must be maintained between 60-65°C throughout.
- Treat 15-25 kg (or a convenient quantity) per batch, depending on vessel size and the quantity of vanilla available.



Figure 33. Pods in a hot water bath

17.1.2 Sweating

The next stage is sweating, during which the treated vanilla pods are kept warm to initiate browning and aroma fixation. The pods are placed in an insulated wooden box for at least 24 hours but less than 48 hours.

The box should be made from well-seasoned, odor-free wood, such as mahogany, with dimensions suited to the quantity of vanilla being processed. Oversized boxes should be avoided because they are difficult to handle. Approximately 1-inch-thick plywood may be used, with a tightly fitting lid made by adjusting approximately ½ inch along the sides and the corresponding edges of the lid to ensure a snug fit. Lining the interior of the box, including the lid, with thick fabric such as flannel or blanket material secured using small tacks helps retain heat and absorb moisture released from the pods.

After hot-water treatment, the pods should be allowed to drain, wrapped in thick cloth, and transferred into the box as quickly as possible to prevent heat loss. They should be packed tightly to fill the box completely and minimize trapped air. The box can then be wrapped externally with blanket material to provide additional insulation. Immature, short, and split pods may be placed in the same box, but each category must be wrapped separately in thick cloth. When adding these pods, the box should be opened briefly, packed promptly, and closed immediately to preserve the internal temperature.

After sweating for more than 24 hours but less than 48 hours, the pods should be removed and sorted again. An internal temperature of approximately 45°C when the box is opened is considered desirable. Well-matured pods may develop a brown colour and an oily-looking surface, although some may not show these changes immediately. Pods remaining dark green may indicate that the box temperature was insufficient, while immature pods may develop a dark grey colour.



Figure 34. Pods laid out for drying

17.2 Rapid Drying Stage - Sun Drying

Thick cotton blanket fabrics are used for this step. A raised platform off the ground, or a wooden rack in a very clean, dry place, is suitable. Minimize the chance of other foreign substances mixing as much as possible. Blanket fabrics must be clean and dry. When vanilla is not being processed, keep these clothes dried in the sun and clean and dry; do not use them for drying other things or other work.

- Pods kept in boxes for close to 48 hours should be sorted by external colour and size.
- Spread in single rows on blankets and dry in strong sunlight for about 3 hours, turning 3 or 4 times.
- Wrap the four corners of the cloth, keep in the sun for another 1-2 hours, pack well, and return to the boxes.

- Repeat this process daily until a good brown colour develops and pods become soft, over about 6-8 days, during which moisture is further reduced to about 40-45% through continued sweating.
- Pods that have not developed good colour or softness are given further days of sun exposure, up to about 20 more days, though pods taking longer are generally lower in quality.

During sudden rain, pods must be moved to shade as quickly as possible; brief light rain can be managed by covering with a polythene sheet or spreading on racks in the processing shed. Several consecutive rainy days can cause mold and reduce quality; on such days, artificial drying in a dryer at 50-60°C for 2-3 hours can be used instead.



Figure 35. Sun-drying vanilla pods

17.3 Slow Drying

Pods are dried on wooden or other racks inside the vanilla processing shed, which must be very clean with good ventilation.

- Racks should start 1-1½ feet above the floor and 1-1½ feet below the roof, with about 6 inches of spacing between racks.
- Pods, arranged in clean trays on the racks, are left to dry slowly for about 2 months.
- Inspect all pods frequently - remove unsuitable, moldy pods, and occasionally rub pods between the fingers.
- Pods dried to the proper extent (25%-30% moisture) are removed from the racks at this point.



Figure 36. Pods in trays during drying

17.4 Conditioning

Pods from the previous step are rubbed between the fingers, allowing secretions to spread over the outer surface, which helps keep the pods straight and gives them a glossy appearance.

- Boxes similar to those used during sweating can be reused, provided they are manageable in size. The largest box should generally hold about 25 kg and should be lined with oiled paper instead of cloth.
- 50 pods of the same size and grade are tied into bundles with black thread at three points (middle and both ends).
- Bundles of the same grade are packed into the same box and left closed to condition for 2-3 months.
- Open the box once a week for thorough inspection, removing any pods showing mold or other signs of deterioration. Pods developing an unpleasant odor due to mold, rot, or pest infestation should be removed immediately.
- Pods releasing excessive water should be re-dried in the shade.
- Remove moldy pods, clean them again, and dry. For this, quick washing in hot water (70 degrees Celsius) and redrying is suitable.

Well-processed pods may sometimes develop very small vanillin crystals on the surface. Following good processing procedures, especially cleanliness of boxes and equipment, produces a good product. If a part of a pod is rotted, that part can be cut off and the remaining part processed separately.

17.5 Quality Control During Curing

- The temperature of the killing bath should never exceed 65°C, and timing is equally crucial.
- Sweating over an extended period causes beans to deteriorate.
- Fungal infection appearing during curing should be addressed immediately - infected beans should be re-killed in 65°C water for a few seconds and the drying procedure repeated until the mold is gone.
- Quality should be evaluated at each stage, including a cross-check on bean weight as beans move through the stages - important for selecting the proper mix and practice.



Figure 37. Fully cured vanilla beans

Characteristics of well-cured beans:

- Capable of wrapping around a finger.
- Colour ranging from dark brown to black.
- Strong aroma and flavor.
- Moisture content of about 30-35% (within the general 25-35% range typical of conditioned vanilla).
- Long, supple, pliable, fleshy beans with a slightly oily appearance.
- Clean beans without scars.

Characteristics of poorly-cured or low-quality beans:

- Reddish-brown to dark-brown colour.
- Dry, hard, thin beans.

- Poor aroma.
- Moisture content around 10%, indicating excessive drying.

18. Other Methods

Apart from the Bourbon method, the Mexican method is also used extensively in vanilla processing elsewhere in the world. It differs from the Bourbon method in that pods are killed by exposure to strong sunlight rather than hot water: pods are left in the shade for 5-7 days and then spread in strong sunlight for about 5 hours to kill them. All other steps follow the same sequence as the Bourbon method.

In South India, pods are killed by boiling, then placed in tied polythene bags on a raised platform in strong sunlight for 3 hours; the next day the bag is untied, water drained, and the process repeated for 3 days, after which the sequence from rapid sun drying onward follows the Bourbon method.

18.1 Comparative International Curing Practices

The following summarizes how curing is traditionally carried out in other major vanilla-producing countries, for reference and comparison; the Bourbon method described in section 17 above remains the recommended procedure for Sri Lankan conditions.

Madagascar and other Bourbon-producing Indian Ocean islands:

Bourbon vanilla originates from a curing process developed on Réunion (formerly Île Bourbon) and is produced across the former French Indian Ocean colonies (Madagascar, Comoros, Mauritius, Seychelles, Réunion, Mayotte, Sri Lanka); Madagascar dominates production of this type. Killing is done by hot-water scorching (63-65°C, 2-3 minutes for the best grades, under 2 minutes for smaller split beans), followed by wrapping in black cloth and 24 hours in a fabric-lined sweating box. Sun drying follows on black cloths over bamboo platforms about 70 cm above ground, 1 hour direct sun then 2 hours wrapped to retain heat, repeated for 6-8 days. Slow drying in the shade on ventilated racks (about 12 cm apart) continues for 2-3 months, with regular sorting; some locations use indoor ovens at 45-50°C in poor weather. Conditioning (straightening, bundling in 50s with black twine, wax-paper lining in metal boxes) takes at least 3 months, with regular inspection and removal of moldy beans. The whole Bourbon curing process takes about 5 to 8 months; Madagascar's main harvest runs from June to early October.

Puerto Rico:

A 1940s variant of the Bourbon method: beans are cleansed with a moist towel, then scalded with three 10-second immersions in an 80°C water bath at 30-second intervals, before being wrapped in a blanket and sweated. Daily two-hour sunnings and nightly sweatings continue for about 7 days until supple, followed by indoor air-drying until the beans shrink to one-third of their weight, then conditioning in tin boxes until they reach one-quarter of their original weight. Because of the higher scalding temperature, Puerto Rican vanilla rarely develops surface frosting.

Mexico:

Two traditional methods are used - sun-wilting (the older method) and oven-wilting (introduced around 1850) - both still used by specialized curing enterprises; the harvest season runs November to January and curing takes about 5 to 6 months. In sun-wilting, sorted beans are killed by about 5 hours in strong sun on mats or racks, then wrapped in blankets and sweated for 12-24 hours in airtight mahogany boxes; beans that fail to darken are oven-wilted instead. This is followed by daily sunnings (2-3 hours) and overnight sweatings for 5-6 days, then less frequent sunning and sweating for 20-30 days (4-8 total sweatings), then about a month of slow indoor drying, and conditioning (bundling of 50, wax paper, metal boxes) for at least 3 months. Oven-wilting uses a brick/cement chamber (calorifico) about 4×4×4 m: blanket-and-matting-wrapped bundles (malletas) are dampened and heated to 60°C by about 12 hours and 70°C by 16 hours, held a further 8 hours, then removed after 36 hours (extended to 48 hours if 65°C cannot be reached), after which sweating, sunning, and conditioning proceed as in sun-wilting.

Tahiti:

Beans are not killed artificially; they are left on the vine until fully mature, indicated by browning of the tip, then collected and cured by specialized companies using a method similar to the Mexican process. The main harvest is February-March, with drying completed by July-August. Fully browned beans are sunned for 3-4 hours on high wooden racks, then sweated overnight in boxes; this is repeated for 15-20 days with progressively fewer night sweatings, after which beans are dried in 10 cm layers in the open breeze. Tahitian vanilla is typically sorted and sold soon after drying, without an extended conditioning period, though beans may be kept in tins if stored longer before sale.

19. Grading Cured Vanilla Pods

Before packing, pods must be graded. The following factors are considered:

- Pod length - best grade: between 17-25 cm.
- Characteristic aroma - inherent, high vanilla aroma.
- Colour and exterior appearance - uniformly spread dark brown or black colour, velvety soft and fleshy, with an oily, glistening nature, free of blemishes and insect infestation.
- Cleanliness - free of all external impurities, animal parts, and insects.
- Moisture - about 25%-30%; when tightly dried, moisture in lower vanilla grades may be as low as 10%.

Good vanilla pods must have more than 2% vanillin content. Grading methods and the number of grades vary across countries - Mexico, for example, uses 7 grades - but pods are generally seen graded into three broad categories: un-split, split, and pieces.

20. Packing

Vanilla pods conditioned in boxes for 2-3 months are packed for market in tins or cardboard boxes. A single package should contain vanilla of only a single grade. Oiled paper is lined inside tins or boxes, and pods are packed in bundles of 50. Boxes should be made airtight; in the case of tins, bundles are packed as a set and then packed again into wooden boxes. Vanilla packed in tins can be kept for a longer time.

21. Uses

Vanilla is mostly used in powder form or as solvent extracts, and is the most popular substance used globally to flavor appetizing foods. It is heavily imported to flavor ice cream, and is also extensively used in chocolate and other confectionery, biscuits and other bakery products, and alcoholic and non-alcoholic beverages. It is used in the perfume industry and, to a limited extent, in the pharmaceutical industry.

22. Vanilla Products

Vanilla-producing countries export vanilla as whole pods or pieces. Consuming countries then prepare other products from it, including:

- Vanilla extract
- Vanilla oleoresin
- Vanilla powder
- Vanilla alcoholic solutions (used in the pharmaceutical industry)

22.1 Vanilla Extract

This is a water-alcohol solution containing the aromatic chemicals present in vanilla; sugar and glycerin are also included to give sweetness and thickness. Traditional vanilla extracts contain at least 85% ethyl alcohol along with dissolved vanilla chemicals.

The quality of a vanilla extract depends on:

- Careful handling and storage of pods before extraction
- Selection and blending of vanilla pods
- Size of pod pieces
- The extraction method and its conditions
- Proper aging of the extract

When pods with about 25-30% moisture are stored at under 10°C and low atmospheric humidity, their characteristic aroma develops further. Two methods are followed for making extract: maceration and continuous recirculation.

Extraction by maceration:

The traditional vanilla extraction method. Crushed pods are placed in vessels, solvent is poured in, and the mixture is left for about a year. Historically, wooden vessels with 50% alcohol were used; today, airtight stainless steel or glass vessels with about 60% alcohol are used, producing a good-quality extract within 1-3 months, after which the solution is slowly removed and filtered. Although this process is long, it yields a better-quality product than continuous recirculation.

Extraction by continuous recirculation:

The method currently used for large-scale vanilla extract production. Perforated trays holding packed vanilla pieces are arranged one above another inside a stainless steel or tin vessel; solvent sprayed onto the topmost tray filters down through each tray to the bottom, and the collected solution is recirculated by pump back onto the top tray. This continuous extraction must run for at least five consecutive days; the pieces are then washed with cold water and the solution centrifuged or filtered. The solution is then aged - desirably for 6 months, but at

least 3 months - to allow the aroma to develop fully, though extract from long-term maceration is better aged. These extracts must be stored in stainless steel, tin, or glass vessels.

22.2 Vanilla Oleoresin

Made by distilling the extract obtained from a maceration-type process applied to vanilla pieces under a single unit. Hot water containing 50% ethanol (by volume), or hot liquid isopropyl alcohol at 50% (by volume), is suitable for this process.

22.3 Vanilla Powder

Made by mixing vanilla pods with sugar or another suitable grinding medium and grinding them together; due to their high moisture and fat content, vanilla pods cannot be ground by themselves.

22.4 Vanilla Alcoholic Solutions

Several vanilla alcoholic solutions exist. Vanilla absolute is made by extracting vanilla with 60%-80% hot ethyl alcohol and then removing the solvent; due to its high resin content it is used only in limited perfume applications. In most perfume industries, vanilla solutions are instead made by macerating vanilla (10 g) in alcohol (100 ml).

23. Market Potential

Vanilla is known as an expensive crop among spices, second only to saffron in the world market.

- Annual natural vanilla production globally is about 5,000 metric tons, of which 90% is supplied by Indonesia and Madagascar.
- 80% of this world production is imported by a few countries - the USA, Germany, and France.
- Vanilla contributes 0.75% of quantitative world spice trade but 7% of its total value.
- Denmark has been identified as the country with the highest per-capita consumption of vanilla.

Annual artificial vanilla production and consumption is about 28,000 metric tons; consumers are increasingly shifting away from artificial vanilla toward natural vanilla, which is healthier. For this reason, high market potential for natural vanilla can be expected in the future.

24. Market Requirements and Pricing of Green Vanilla Pods

The market price of green (raw) vanilla pods in Sri Lanka is generally determined by their length, maturity, and overall quality. Green vanilla pods may typically be purchased at approximately Rs. 5,000 - 7,000 per kilogram, depending on their quality. However, these prices are indicative and may vary according to market conditions and buyer requirements.

Pod length is an important factor in determining market value. Pods measuring 7 inches or more are generally preferred for obtaining better prices. Mature, healthy, well-developed pods with minimal physical damage are more desirable in the market.

Immature, split, diseased, insect-damaged, severely stained, or mold-affected pods may receive lower prices or be rejected by buyers. Therefore, harvesting pods at the appropriate maturity stage and maintaining their quality are important for obtaining satisfactory market returns.

25. Technical Advice and Planting Materials for Vanilla Cultivation

Technical advice, training programs, and planting materials for vanilla cultivation can be obtained from the offices of the Department of Export Agriculture and the In-Service Training Centre located at Elwala, Matale. For further details, contact the Assistant Director of Export Agriculture or the Extension Officer of your area.

District Offices and Contact Directory - Department of Export Agriculture

District / Unit	Address	Telephone Number
Kurunegala	10, Wilgoda Road, Kurunegala	037-2222160
Matale	Elwala, Ukuwela	066-2243451
Kandy	1060, Nuwara Eliya Road, Peradeniya	081-2388392
Nuwara Eliya	Racecourse Road, Nuwara Eliya	052-2222065

Badulla	Ratwatte Mawatha, Badulla	055-2222057
Ratnapura	New Town, Ratnapura	045-2222029
Kegalle	Rathambalawatta Road, Kegalle	035-2222204
Gampaha	313/64/09, Walawwatta, Colombo Road, Gampaha	033-2222553
Colombo & Kalutara	Fruit Factory Road, Colombo 05	011-2508725
Galle	Bandipora, Labuduwa, Akmeemana, Galle	091-2223494
Matara	38/A, Rahula Mawatha, Matara	041-2222443
Hambantota	Kadurupokuna Road, Tangalle	047-2240214
Monaragala	Kachcheri, Monaragala	055-2276068
Head Office	1095, Peradeniya	081-2388651
Information Unit	Dharmawansa Bandaranaike Mawatha, Peradeniya	081-2388018 / 081-2388019
Training Institute	Elwala, Ukuwela	066-2243450
Plant Protection	Elwala, Ukuwela	066-2243457
Cinnamon Research Station	Palolpitiya, Kihagoda	041-2245407

26. Growing Vanilla in Greenhouses

The goal of greenhouse vanilla growing is to reproduce the conditions found beneath the rainforest canopy. To accomplish this, climatic parameters such as temperature and humidity are managed with equipment such as fogging systems and shade screens. The greenhouse also protects vanilla from climate risks such as excess water, which can lead to disease

outbreaks, and allows for year-round human interaction and regular monitoring of the orchid's progress.



Figure 38. Vanilla nursery inside a polytunnel

Benefits of growing vanilla in a greenhouse include:

- Improved climate management inside the growing area, allowing precise monitoring and adjustment of relative humidity, temperature, and light transmission.
- An effective ventilation system to remove excess humidity and recycle air as needed.
- Higher production density within the glasshouse (around 20 plants/m²) due to improved air circulation and ergonomics.
- Maximum growth of up to 25 cm per week, with monthly flowering achievable.
- Improved pest and disease protection through integrated pest management in the growing area.

27. Quality Control and Storage of Cured Vanilla Beans

Proper storage following the gradual conditioning process is also crucial for final quality, since how vanilla beans are kept and maintained will influence their ultimate scent. Handling at

various stages also requires specialized trays and material-handling equipment appropriate to the process.

- Suitable phytosanitary precautions should be taken to prevent pests and diseases in storage, including molds, fungi, and scale or mite infestations.
- The storage space should be clean and well-kept, with suitable aeration and humidity control.
- All wrapping materials should be sterile.
- Contaminated beans should be separated from the rest of the batch.
- Excessive drying causes moisture loss and makes beans more brittle over time.
- Most beans are vacuum-packed, using appropriate vacuum-packing methods and poly liners, to eliminate air-related problems and preserve the beans intact for longer.



Figure 39. Bundled beans packed for storage

28. Vanillin in Vanilla

Natural vanillin is derived from the seed pods of Vanilla planifolia, a vining orchid native to Mexico now grown in tropical locations worldwide; Madagascar is currently the world's largest producer of natural vanillin. The green seed pods contain vanillin in the form of its glucoside (glucovanillin) when harvested - the green pods themselves have no vanilla flavor or odor.

After picking, flavor develops through a months-long curing process whose details differ across vanilla-producing countries, but which generally follows this pattern:

1. The seed pods are first blanched in hot water to halt the activity of live plant tissue.
2. The pods are then alternately sunned and sweated for 1-2 weeks: spread in the sun by day and covered in fabric in airtight boxes to sweat at night.
3. The pods turn dark brown during this process as enzymes release vanillin as a free molecule.
4. Finally, the pods are dried and stored for many months to allow their flavor to develop further.

Faster curing methods, measured in days rather than months, have been described but have not been widely adopted in the natural vanilla industry, which prioritizes an established process that preserves the premium flavor profile over speed.

28.1 The Chemistry of Vanillin

Vanillin is a chemical compound with the formula $C_8H_8O_3$, a type of phenolic aldehyde whose functional groups include aldehyde, hydroxyl, and ether groups; it is the major flavor component of vanilla bean extract. Synthetic vanillin is now used more frequently than genuine vanilla extract as a flavoring ingredient in foods, beverages, and medicines. The food industry uses both vanillin and ethylvanillin; ethylvanillin is more costly but has a stronger flavor, differing from vanillin in having an ethoxy group (OCH_2CH_3) rather than a methoxy group (OCH_3). Natural vanilla extract contains hundreds of distinct chemicals in addition to vanillin, whereas artificial vanilla flavoring is typically a solution of pure (usually synthetic) vanillin. Because genuine vanilla extract is scarce and expensive, synthetic production of vanillin has long been of commercial interest; the first commercially viable synthesis began from the more widely available natural compound eugenol (4-allyl-2-methoxyphenol), while today artificial vanillin is manufactured from either guaiacol or lignin.

29. Health Benefits and Other Uses of Vanilla Products

Vanilla extract and other vanilla bean products, such as vanilla paste and powder, are popular flavoring agents, and research suggests that certain chemicals in vanilla extract and beans may have antioxidant, anti-inflammatory, and neuroprotective qualities. However, no studies have yet determined whether these effects apply to people consuming pure vanilla in ordinary dietary amounts, so it remains unknown whether vanilla products have a significant impact on human health.

Simple everyday uses of vanilla products include:

- A pinch of vanilla bean powder or extract in coffee.
- A dash of powder added to smoothies.
- Vanilla extract or A-grade beans used in baking cakes or cookies.
- Vanilla paste or powder added to waffle and pancake mixes.
- Pure, grade-A vanilla beans used in ice cream recipes.

29.1 Anti-inflammatory Properties

Vanillin has been shown in animal and test-tube studies to have anti-inflammatory properties. Persistent inflammation is linked to a number of health issues, including increased risk of chronic illness, and diets rich in anti-inflammatory foods may lower inflammatory markers and improve general health. In a 2018 study, mice fed a high-fat diet to induce obesity showed lower levels of inflammatory markers (interleukin-6 and tumor necrosis factor-alpha) in blood and liver tissue after oral vanillin supplementation; a 2017 study found oral vanillin reduced psoriasis-related skin inflammation in mice; and other rodent and test-tube studies have reported protection against inflammation-related cell damage. However, there is no human evidence that ingesting vanilla extract or beans reduces inflammatory markers or protects cells from inflammation-related damage - the compounds used in these studies were concentrated extracts, not whole vanilla products.

29.2 Antioxidant Properties

Some molecules in vanilla extract and beans, notably the phenolic compounds vanillin and vanillic acid, have antioxidant properties that help protect cells from damage. A 2020 test-tube study found both compounds protected brain cells from oxidative damage, with vanillin the more potent of the two; studies in aged rats found vanillin protected against liver damage and age-related oxidative damage, and a 2011 rat study found pretreatment with vanillin

protected against induced liver damage. While vanilla extract and beans clearly contain antioxidant-protective compounds, it remains uncertain whether regular dietary doses provide antioxidant benefits to humans.

29.3 Brain Health (Neuroprotective Properties)

Some studies suggest vanillin and vanillic acid may benefit brain function and have neuroprotective qualities. A 2021 study examined the neuroprotective effect of vanillic acid in mice injected with a neurotoxin, finding that vanillic acid injections reduced nerve-cell inflammation, decreased markers associated with Alzheimer's disease, and reduced neurotoxin-induced memory impairment; vanillin has similarly been identified as a potential neuroprotective agent in other studies. However, there is no human evidence that consuming vanilla extract or beans improves brain function or protects against cognitive decline.

29.4 Reducing Added Sugar Intake

Using vanilla extract or powder in dishes and beverages may help reduce added sugar intake. A 2020 study of 129 young adults found that adding vanilla scent to sugary drinks increased their perceived sweetness; a 2021 study found flavoring low-sugar yoghurt with vanilla had no effect on perceived sweetness; and a separate 2020 study of 112 children found that adding vanilla to milk-based sweets increased overall liking of reduced-sugar alternatives for more than 80% of participants, suggesting vanilla may help lessen the need for extra sugar in some products.

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